**EXP 7: Caching Application Data with ElastiCache, Caching with Amazon CloudFronT, Caching Strategies**

## Theory:

AWS CloudFront - CloudFront offers several options for streaming your media to global viewers—both pre-recorded files and live events. For video on demand (VOD) streaming, you can use CloudFront to stream in common formats such as MPEG DASH, Apple HLS, Microsoft Smooth Streaming, and CMAF, to any device.

The two main components of AWS Cloudfront are content delivery and dynamic content caching

**Step to execute:**

**Step 1: Create Redis Cache (ElastiCache)**

- Go to AWS Console.
- On the left side, select **Resources → Redis OSS Cache**.
- Click **Continue to Redis OSS**.
- Configure the settings:
    - Disable **Multi-AZ**
    - Select **Engine version: 7.0**
    - Set **Port number: 6379**
    - Choose **Node type: t3.micro**
    - Set **Shards: 2**
- Provide a **Redis subnet group name** and proceed.

**Step 2: Configure Authentication**

- Select **Auth default access**.
- Set an **Auth token** (example: Authtoken123456789).

**Step 3: Create Security Group (EC2)**

- Open a new tab → Go to **EC2 Dashboard**.
- Create a **Security Group**.
- Add **Inbound Rule**:
- Type: Custom TCP
- Port: **6379**
- Source: **Anywhere (IPv4)**
- Click **Create Security Group**.

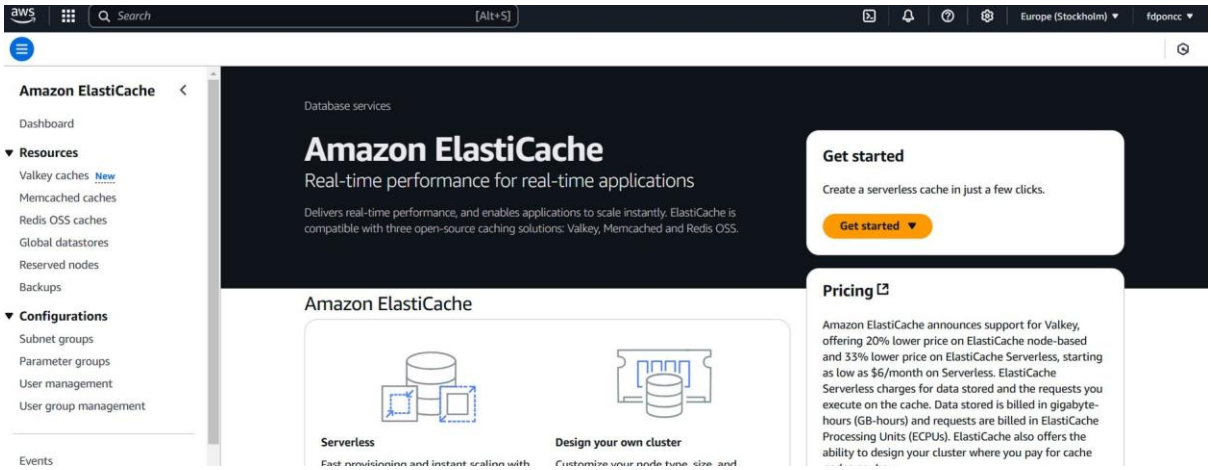
**Step 4: Attach Security Group to ElastiCache**

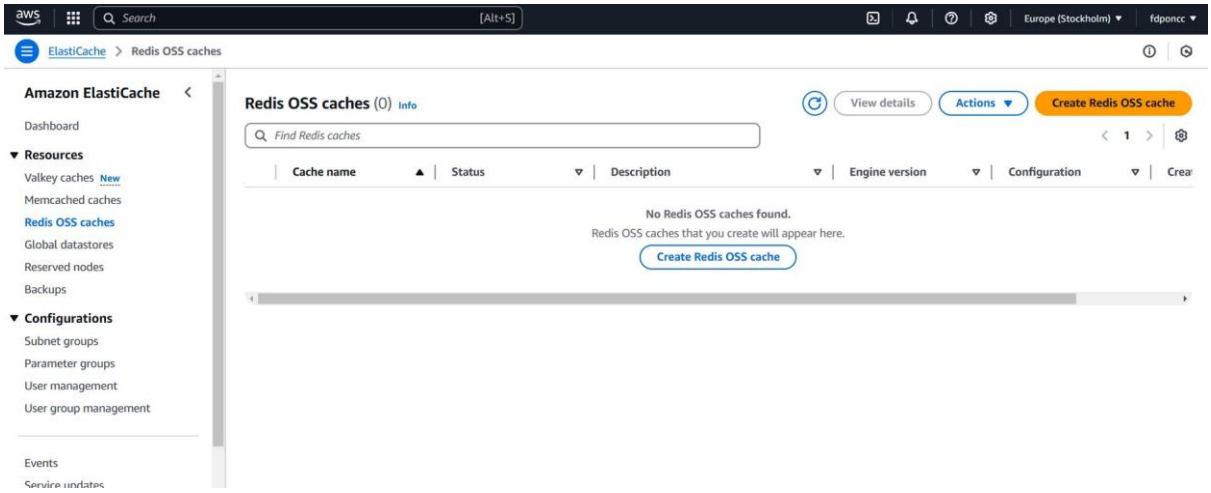
- Go back to **ElastiCache**.
- Select the created Redis cluster.
- Go to **Manage → Refresh**.
- Attach the newly created **Security Group**.

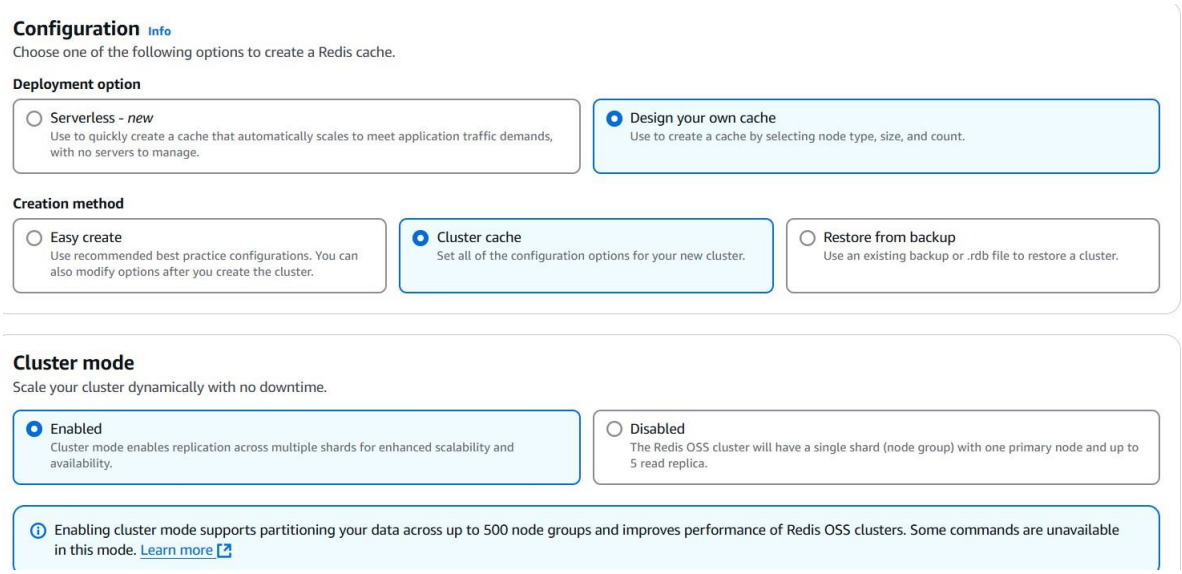
**Step 5: Create Cache**

- Proceed with remaining default settings.
- Click **Create** to launch the Redis cache.

**EXP 7: Caching Application Data with ElastiCache, Caching with Amazon CloudFronT, Caching Strategies**



## Cluster settings

Use the following options to configure the cluster.

**Engine version**

Version compatibility of the engine that will run on your nodes.

7.0

**Port**

The port number that nodes accept connections on.

6379

**Parameter groups**

Parameter groups control the runtime properties of your nodes and clusters.

default.redis7.cluster.on

**Node type**

The type of node to be deployed and its associated memory size.

cache.t3.micro
0.5 GiB memory  Up to 5 Gigabit network performance

**Number of shards**

Enter the number of shards in this cluster, from 1 to 500.

2

**Replicas per shard**

Enter the number of replicas for each shard, from 0 to 5.

1

## Connectivity

Choose the IP version(s) this cluster will support. Then select an existing subnet group or create a new one.

**Network type**

Choose between IPv4, dual stack and IPv6

IPv4
Your resources will communicate only over the IPv4 protocol.

**Subnet groups**

Choose existing subnet group | Create a new subnet group

**Name**

MYREDDIS26-SUBNET

The name can have up to 255 characters, and must not contain spaces.

**Description - *optional***

*Description*

**Subnet groups**

Choose existing subnet group | Create a new subnet group

**Name**

MYREDDIS26-SUBNET

The name can have up to 255 characters, and must not contain spaces.

**Description - *optional***

*Description*

**VPC ID**

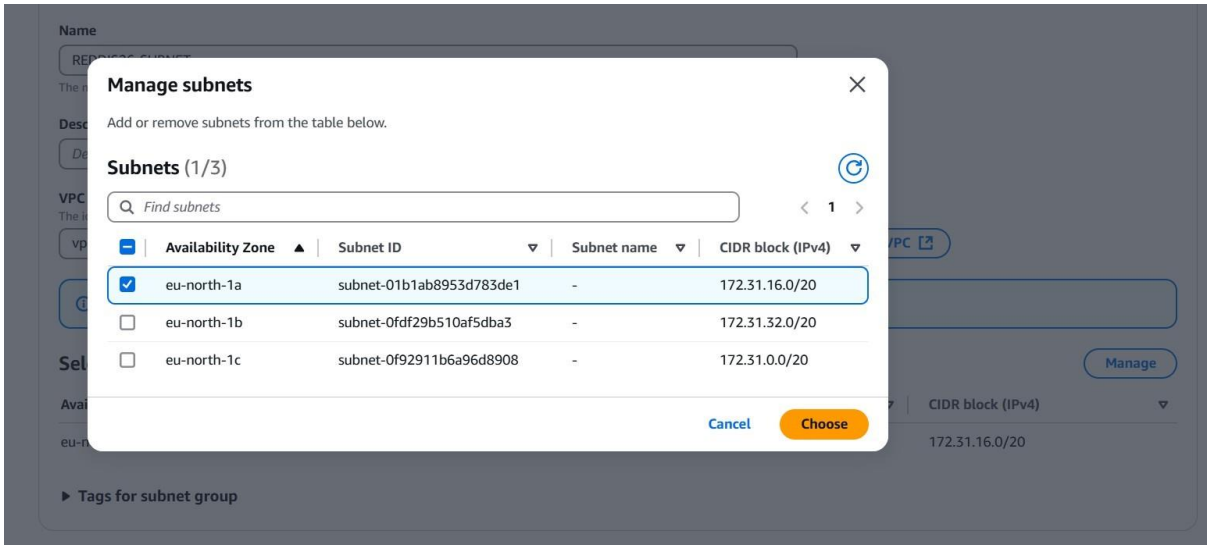
The identifier for the VPC environment where your cluster is to run.

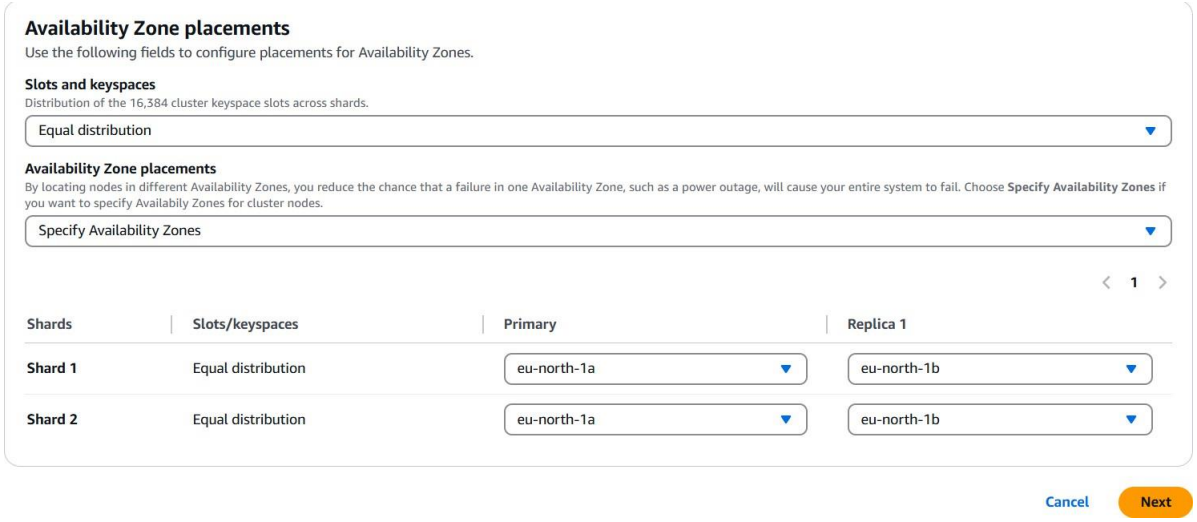
vpc-0dc4bbbdcb80c11cb  Create VPC

For Multi-AZ high availability mode, choose IDs for at least two subnets from two Availability Zones in the table below.

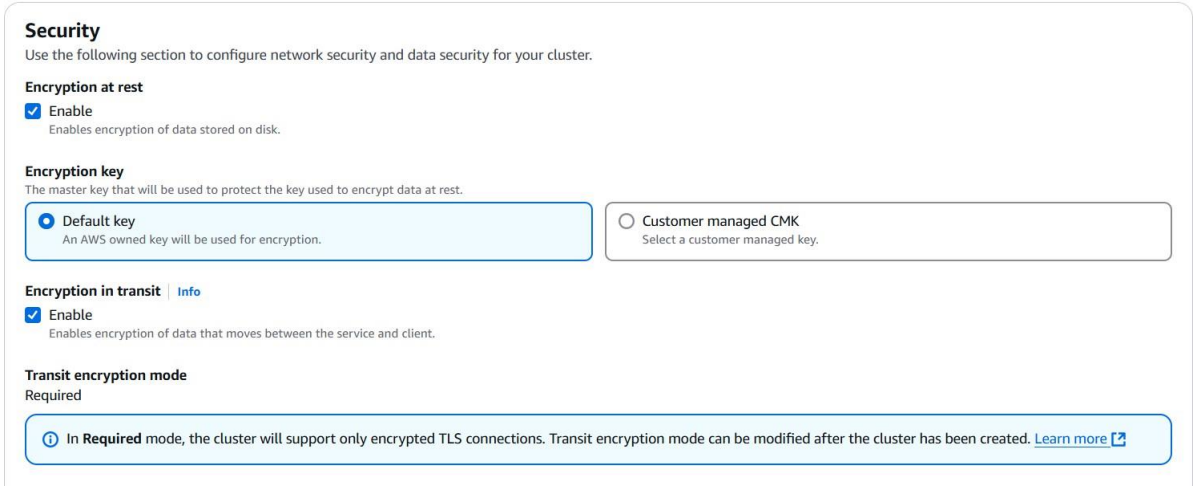
**Selected subnets** (2)  Manage

| Availability Zone | Subnet ID | Subnet name | CIDR block (IPv4) |
|---|---|---|---|
| eu-north-1a | subnet-01b1ab8953d783de1 | - | 172.31.16.0/20 |
| eu-north-1b | subnet-0fdf29b510af5dba3 | - | 172.31.32.0/20 |




## Advanced settings Info



**Access control**

Provides the ability to configure authenticating and authorizing access.

AUTH default user access

**AUTH token**

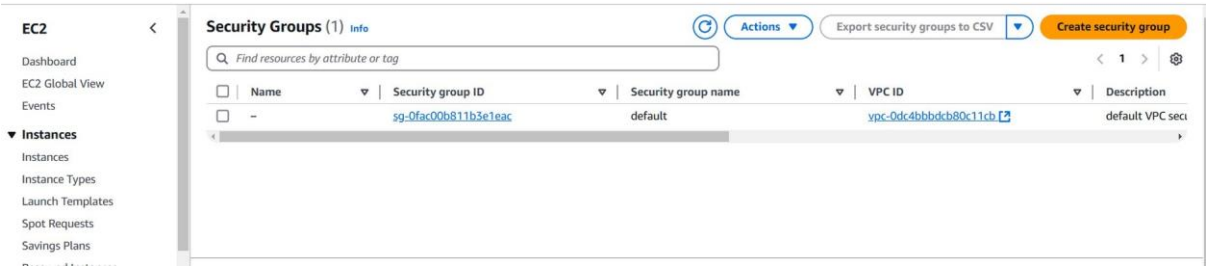
The AUTH token used for the cluster.

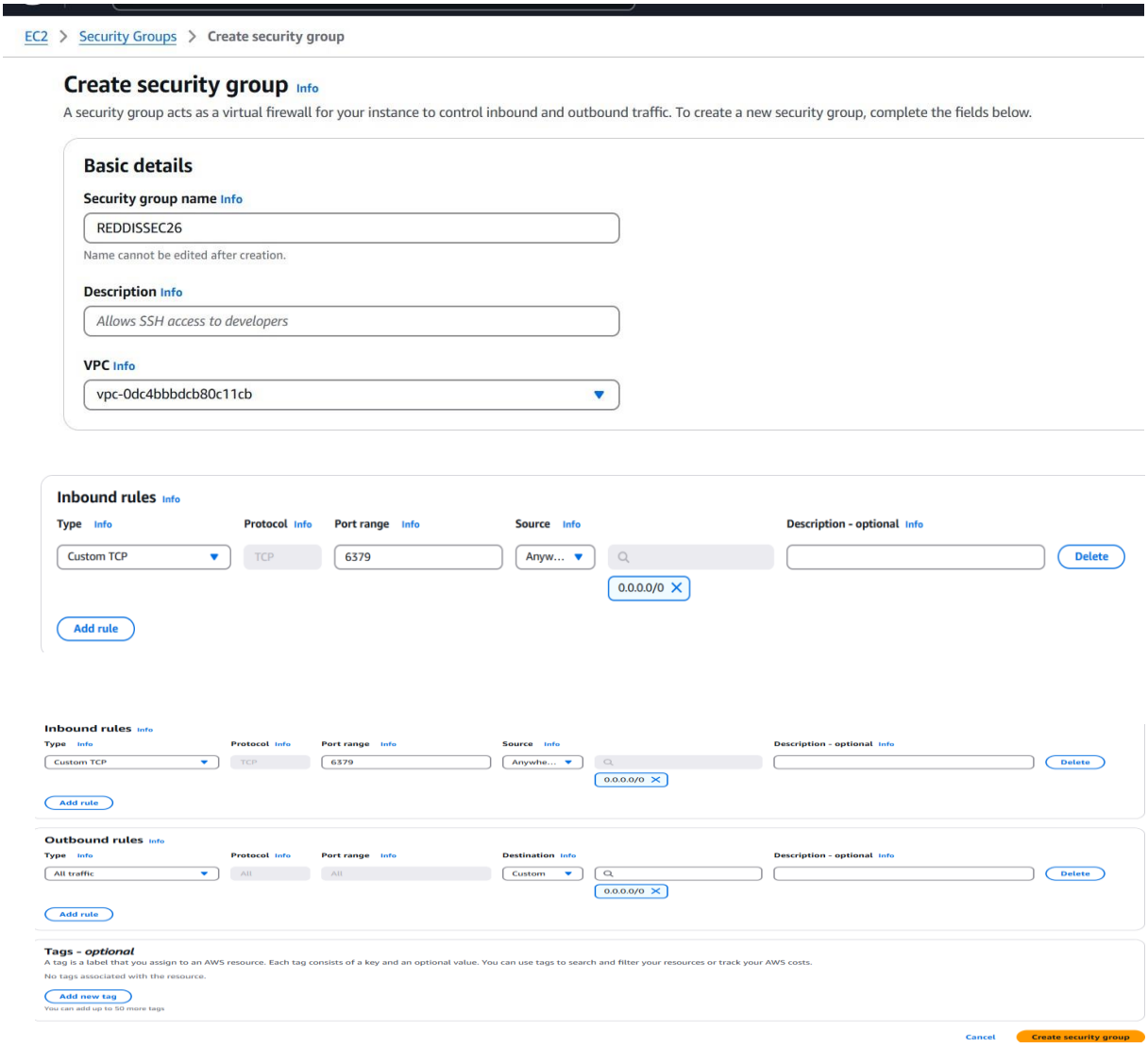
AUTHtoken123456789

☑ Show token

At least 16 characters and a maximum of 128 characters, which can be any printable ASCII character except for ' ' (blank space), '"' (quotation mark), '/', and '@'.

## GO TO EC2 IN NEW TAB AND CREATE A SECURITY GROUP



Security group (sg-08b8fd734ba812a46 | REDDISSEC26) was created successfully
Details

## sg-08b8fd734ba812a46 - REDDISSEC26

Actions

### Details

| Security group name | Security group ID | Description | VPC ID |
|---|---|---|---|
| REDDISSEC26 | sg-08b8fd734ba812a46 | REDDISSEC26 | vpc-0dc4bbbdcb80c11cb |

| Owner | Inbound rules count | Outbound rules count |
|---|---|---|
| 699475937818 | 1 Permission entry | 1 Permission entry |

Inbound rules | Outbound rules | Sharing - *new* | VPC associations - *new* | Tags

### Inbound rules (1)

Manage tags | Edit inbound rules

Search

| Name | Security group rule ID | IP version | Type | Protocol | Port range | Source | Description |
|---|---|---|---|---|---|---|---|
| – | sgr-05bc2e2303a61e4a0 | IPv4 | Custom TCP | TCP | 6379 | 0.0.0.0/0 | – |

### Selected security groups (0)

Manage

A security group acts like a firewall that controls network access to your clusters.

| Group ID | Name |
|---|---|

No selected security groups

Add security groups by choosing the Manage button.

Manage

AUTH default user access

### Manage security groups

### Security groups (1/2)

Find security groups

| Security group ID | Security group name | Description |
|---|---|---|
| sg-08b8fd734ba812a46 | REDDISSEC26 | REDDISSEC26 |
| sg-0fac00b811b3e1eac | default | default VPC security group |

Cancel | Choose

### Backup

You can use backups to restore a cluster or seed a new cluster. The backup consists of the cluster's metadata, along with all of the data in the cluster.

☐ Enable automatic backups
ElastiCache will automatically create a daily backup of a set of replicas.

### Maintenance

Configure maintenance settings for the cluster.

**Maintenance window**
Specify the time range (UTC) for updates such as patching an operating system, updating drivers, and installing software or patches.

◉ No preference
○ Specify maintenance window

**Auto upgrade minor versions**

☐ Enable
Automatically schedule cluster upgrade to the latest minor version, once it becomes available. Cluster upgrade will only be scheduled during the maintenance window.

**Topic for Amazon SNS notification**
Choose an SNS topic from the list, or enter the Amazon Resource Name (ARN) for an existing topic. If no topic is chosen, no notifications are sent.

Disable notifications

## Logs

Specify whether to provide the Redis OSS slow logs or engine logs.

**Slow logs**

☑ Enable

Provide the slow log for queries that exceed a specified runtime.

**Log format**

Choose the format of the logs.

Text

**Log destination type**

Choose the destination of the logs.

CloudWatch Logs

**Log destination**

Select an existing log group or create a new one.

○ Choose existing log group | ◉ Create a new log group

**Log group name**

REDDIS26LOG

**Engine logs**

☐ Enable

Provide the engine log for queries that exceed a specified runtime.

## Tags

You can use tags to search and filter your clusters, or track your AWS costs.

No tags associated with the cluster.

Add new tag

You can add 50 more tags.

Cancel | Previous | Next

## Logs

Specify whether to provide the Redis OSS slow logs or engine logs.

| | | |
|---|---|---|
| **Slow logs**<br>Enabled | **Slow logs format**<br>text | **Slow log destination type**<br>cloudwatch-logs |
| **Slow log destination**<br>REDDIS26LOG | **Engine logs**<br>Disabled | |

## Tags

You can use tags to search and filter your clusters, or track your AWS costs.

| Key | Value |
|---|---|

No tags found.

Tags that you create will appear here.

Cancel | Previous | Create

ElastiCache > Redis OSS caches

The cluster was created successfully.

**Redis OSS caches** (1) Info

View details | Actions | Create Redis OSS cache

Find Redis caches

1

| Cache name | Status | Description | Engine version | Configuration | Created date |
|---|---|---|---|---|---|
| myreddis26 | Creating | - | 7.0.7 | cache.t3.micro | February 5, 2025, 03:5... |